JIANXUAN SHAO

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Portfolio mars-1010.github.io/Umang/

RESEARCH INTERESTS

Human-AI Collaboration | Personal AI | Adaptive Decision Support | Spatial Interaction

I aim to develop assistants that share an understanding of the environment, coordinate actions with people and adapt to changing tasks. My preparation combines architectural design, immersive installations, interactive prototyping and machine learning.

EDUCATION

University of Sydney Feb 2025 - Present

Master of Computer Science | Focus: Data Science and AI | Expected Dec 2026

Semester 1, 2026 average: 84.5/100; average across 12 graded units: 79.33/100. High Distinctions: Introduction to Cybersecurity 92; Machine Learning and Data Mining 89; Introduction to Programming 87; Computer and Network Organisation 85.

Hebei University Sep 2018 - Jun 2023

Bachelor of Engineering in Architecture | GPA 4.05/5.0

Five-year programme spanning architectural design, building physics, computational design and digital visualisation; experience interpreting spatial relationships through drawings, models and site work.

RESEARCH AND INTERACTIVE SYSTEMS

Game Fellows 2025 - Present

Independent prototyping, followed by academic guidance | Python / Pygame

- Developed hand-authored following and fetching prototypes, then extended a Galaga-style game with a visible companion under academic guidance. Implemented formation, projectile interception, protection and coordinated support fire.
- Explored a separate screenshot-to-AI route: capture the game screen, interpret the situation with AI and use the interpretation to inform companion responses. The current reproducible study build uses local game-state rules.
- Implemented bounded movement, stable state transitions, cooldowns and action-linked text/audio cues; added counterbalanced Solo/Fellow conditions, paired random seeds and event logging to support controlled comparisons.
- Developed a research manuscript under academic guidance and prepared experience questionnaires covering teammate presence, trust, usefulness and player control. Human-participant evaluation is planned.

Prototype videos and development

Explainable AI-Generated Image Detection 2026 - Present

Direct-detection baseline and error analysis | Eight-person University of Sydney capstone team

- Owned direct-detection baseline verification and reproducibility checks, and co-developed project aims and questions. Audited 60,000 predictions from 10 model runs against the canonical manifest, reconstructing parsed labels with zero mismatches.
- Analysed a team system combining Detection-first and Evidence-first Qwen3.5-9B reasoning with a separate ResNet-50 visual-forensics branch. On 6,000 balanced FakeBench images from 10 generators, team fusion reached 94.33% balanced accuracy on a 900-image internal split, 5.67 percentage points above ResNet-50; model screening had included this split.
- Independently reconstructed all 900 fused decisions from frozen outputs and coefficients, categorised the 51 errors and identified class-specific costs hidden by aggregate improvement. Prepared reproducible case analyses and the final-report limitations section.

SPATIAL SYSTEMS AND ARCHITECTURAL EXPERIENCE

CAVE Immersive System and Cultural Tourism
Jul 2023 - Jul 2024

Interactive Media Intern | Shanghai Fengyuzhu Culture Technology Co., Ltd.

- Worked on the 2023 Damo Cave immersive exhibition: spatial layout, lighting placement and tuning, projection content and equipment configuration, and on-site integration.
- Participated in setting up and debugging an immersive system combining multi-channel projection blending, wall-touch control and radar recognition, checking display and interaction components together on site.
- Applied VR/AR visualisation and digital storytelling to architectural concepts in the "Dragon Roaming by the Riverside" cultural tourism project.

Daci Pavilion and Geyuan Temple
Oct 2021

Survey and reconstruction team | Hebei University

- Recorded historic buildings through field sketches and CAD/Rhino drawings; reconstructed plans, elevations, sections and three-dimensional geometry, including traditional timber bracket components and architectural details.

Damo Cave Browser Interaction Exploration
2026

AI-assisted portfolio prototype | JavaScript / Canvas 2D / MediaPipe

- Developed a subsequent browser exploration connecting hand/body input with projection brightness, scene reveals and movement-driven floor effects, with mouse fallback. This photographic interaction prototype is separate from the 2023 installation.

Browser interaction demo

SELECTED COMPUTATIONAL PROJECTS

Classification and PathMNIST Pipelines
2026

Machine Learning and Data Mining coursework

Built preprocessing, stratified cross-validation and tuning pipelines; compared Random Forest, MLP and CNN models using macro-F1, class-level errors and runtime.

Electricity Data Engineering and Spatial Integration
2026

Data Engineering coursework

Integrated geocoded energy and regional datasets in spatial DuckDB; processed 716,454 facility records and streamed JSON via MQTT to a Streamlit map dashboard.

TECHNICAL AND RESEARCH SKILLS

Programming and data: Python, SQL, Pygame; preprocessing, model evaluation and reproducible analysis.

Interaction and spatial design: State-based behaviour, event logging, CAD, Rhino/Grasshopper, 3D modelling, projection and lighting integration.

Research practice: Prototype iteration, study preparation, error analysis and technical reporting. Browser development with JavaScript, Canvas and MediaPipe is AI-assisted.

ADDITIONAL EXPERIENCE AND AWARDS

Hebei Haoyu, Beidaihe | Architectural Design Intern | Jun-Aug 2022: school layout studies and drawings. Qinhuangdao Guangjian | Assistant Architect Intern | Jun-Aug 2021: redevelopment drawings.

Hebei University Scholarship: Second Prize (2019), Third Prize (2021, 2022); JYZC Innovation & Creativity Contest Nomination (2022). Tennis Club President (2019-2020); campus architectural sketching team leader (2022).